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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Applying Artificial Intelligence to improve on working and living standards

AI is now used in farming, health, banking, education and government. This chapter revisits AI in more depth, its ethics, techniques, machine learning and development, and introduces the fundamentals of computer networks on which AI services depend.

Lesson	Title (MINESEC progression)
1	The concept of artificial intelligence
2	AI Ethics and Responsible Use
3	AI Techniques and Intelligent Systems
4	Machine Learning
5	Developing AI Systems
6	Integration activities
7	Fundamentals of computer Networks

Lesson 1: The concept of artificial intelligence

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ explain the concept of artificial intelligence
- ✔ describe how AI improves working and living standards

KEY POINTS

Artificial intelligence (AI) is the branch of computing that builds systems able to perform tasks that normally require human intelligence: perceiving, learning, reasoning, solving problems, understanding language and making decisions.

Component of intelligence	AI capability	Example
Learning	Improving from data and experience	Spam filter that gets better over time
Reasoning	Drawing conclusions from rules and facts	Expert system diagnosing a fault
Perception	Interpreting images, sounds and sensor data	Face recognition, speech recognition
Language	Understanding and producing human language	Chatbots, translation
Problem solving and planning	Finding the best sequence of actions	Route planning, game playing

AI improving working and living standards

Sector	Application in Cameroon and beyond
Agriculture	Disease detection from leaf photos; weather-based planting advice; drones for spraying
Health	AI reading X-rays and scans; chatbots for health information; predicting epidemics
Finance	Fraud detection in mobile money; credit scoring for small businesses
Education	AI tutors; automatic marking; personalised revision
Transport	Route optimisation for deliveries; traffic management
Government	Processing documents; detecting tax fraud; e-services chatbots
Business	Customer-service chatbots; sales forecasting; marketing
Accessibility	Speech-to-text, text-to-speech, translation into local languages

KEY POINTS

Narrow AI (performs one task) is the only AI that exists today; **general AI** (human-level at all tasks) and **super AI** are not yet achieved.

Example: Explain two ways AI can improve the living standards of farmers.

AI apps can identify crop diseases from photos and recommend treatment early, reducing losses; AI weather and market-price forecasts help farmers plant and sell at the best time, increasing income.

Practice Questions: Lesson 1

1. Give one use of AI in health and one in finance.
2. Which type of AI exists today?

Answers: *1. Health: reading X-rays; finance: fraud detection in mobile money 2. Narrow AI*

Lesson 2: AI Ethics and Responsible Use

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply ethical principles to AI systems
- ✔ identify misuse of AI

Lesson notes

Principle	Meaning	Example of a violation
Fairness	No discrimination against groups	A credit model refusing most women
Transparency and explainability	Decisions can be understood and explained	A loan refused with no reason
Privacy	Personal data collected with consent and protected	An app selling users' location data
Accountability	People remain responsible for AI decisions	"The computer decided" used as an excuse
Safety and security	The AI is reliable and protected from attacks	A self-driving car fooled by stickers on a stop sign
Human oversight	Humans can review and override	Automatic dismissal of workers by an algorithm
Sustainability	Consider energy use and environmental impact	Training huge models with fossil-fuel electricity

KEY POINTS

Bias enters AI through unrepresentative or prejudiced **training data**, poorly chosen features or careless use. It is reduced by diverse data, testing results for different groups, and human review.

Misuse	Risk
Deepfakes	Fraud, blackmail, false political news
Mass surveillance	Loss of privacy and freedom
Automated weapons	Loss of human control over life-and-death decisions
AI-generated misinformation	Public confusion, damage to reputations
Over-reliance	People stop checking AI answers that may be wrong ("hallucinations")

Worked examples

Example: A company uses an AI camera to monitor employees' facial expressions all day. Discuss two ethical concerns.

> **Privacy:** constant monitoring of faces is intrusive and may be done without proper consent. **Fairness and accuracy:** the system may misread expressions (especially for some groups) and lead to unfair treatment of workers.

Practice Questions: Lesson 2

1. State two ways of reducing bias in an AI system.
2. What is an AI hallucination?

Answers: *1. Use diverse, representative data; test results for different groups (human review) 2. A confident but false answer produced by AI*

Lesson 3: AI Techniques and Intelligent Systems

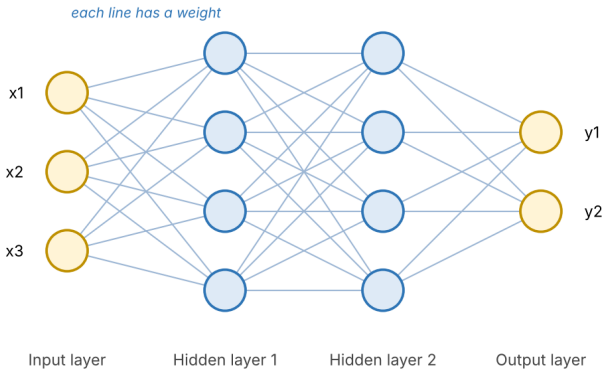
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe AI techniques and intelligent systems

Lesson notes

Technique	Principle	Typical use
Rule-based expert systems	Knowledge base of IF–THEN rules + inference engine	Diagnosis, configuration, advice
Search algorithms	Explore possible states to reach a goal (breadth-first, depth-first, A*)	Navigation, puzzles, games
Machine learning	Learn patterns from data	Prediction, classification
Artificial neural networks	Layers of neurons with learned weights	Images, speech, language
Natural language processing	Process and generate human language	Chatbots, translation, summarising
Computer vision	Analyse images and video	Quality control, face recognition
Fuzzy logic	Reason with degrees of truth (0 to 1)	Appliance control
Genetic algorithms	Evolve solutions by selection, crossover and mutation	Timetabling, optimisation



KEY POINTS

An **intelligent agent** perceives its environment with **sensors**, decides using a program, and acts with **actuators** to achieve goals. Examples: robot vacuum, smart thermostat, trading bots, virtual assistants.

Forward chaining starts from facts and applies rules to reach conclusions; **backward chaining** starts from a hypothesis and searches for supporting facts.

Worked examples

Example: Rules: R1: IF temperature > 38 AND cough THEN suspect flu. R2: IF suspect flu THEN recommend rest and fluids. Facts: temperature = 39, cough. What does forward chaining conclude?

R1 fires (39 > 38 and cough): "suspect flu". Then R2 fires: "recommend rest and fluids".

Practice Questions: Lesson 3

1. Which technique evolves solutions by selection, crossover and mutation?
2. Name the two parts that let an intelligent agent sense and act.

Answers: *1. Genetic algorithms 2. Sensors and actuators*

Lesson 4: Machine Learning

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe types of machine learning
- ✔ evaluate a model's accuracy

Lesson notes

KEY POINTS

Machine learning builds models that learn patterns from data and make predictions on new data without being explicitly programmed with rules.

Type	Learns from	Task	Example
Supervised	Labelled data	Classification, regression	Predicting house prices; detecting fraudulent transfers
Unsupervised	Unlabelled data	Clustering, association	Grouping customers; product recommendations
Reinforcement	Rewards and penalties	Sequential decisions	Robots, game agents, traffic-light control

Term	Meaning
Feature	An input variable (e.g. size, location)
Label	The correct output (e.g. price)
Training set / test set	Data to learn from / unseen data to evaluate
Overfitting	Model memorises training data; performs poorly on new data
Underfitting	Model too simple to capture the pattern
Accuracy	Correct predictions ÷ total predictions
Confusion matrix	Table of correct and incorrect predictions per class

Worked examples

Example: A malaria test model is checked on 500 blood images: 230 true positives, 240 true negatives, 20 false positives and 10 false negatives. Find the accuracy.

Correct = 230 + 240 = 470; accuracy = 470 ÷ 500 = **94%**.

EXAM TIP

In medicine, false negatives (missing a sick patient) can be more dangerous than false positives, so accuracy alone is not enough.

Practice Questions: Lesson 4

1. Which type of learning would group customers with no labels?
2. A model is correct 276 times out of 300. Find its accuracy.

Answers: *1. Unsupervised learning (clustering) 2. 92%*

Lesson 5: Developing AI Systems

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ outline the development of an AI system
- ✔ identify tools used to build AI systems

Lesson notes

1. **Problem definition:** the goal and how success is measured.
2. **Data collection:** sensors, surveys, records, open datasets; obtain consent.
3. **Data preparation:** clean, label, balance, and split (e.g. 70/15/15 for training, validation and testing).
4. **Model selection and training.**
5. **Evaluation:** accuracy and other measures on test data; check for bias.
6. **Deployment:** mobile app, website, embedded device or cloud service.
7. **Monitoring and maintenance:** retrain as data changes; collect feedback.

KEY POINTS

Garbage in, garbage out: data must be sufficient, accurate, relevant, representative and legally collected.

Tool / platform	Use
Python (NumPy, pandas, scikit-learn)	Data handling and classic ML
TensorFlow, PyTorch	Deep learning
Google Colab, Jupyter	Running code in notebooks
Teachable Machine	Training image/sound models without code
Cloud AI services (APIs)	Ready-made vision, speech and language models

```
# Predicting a student's final mark from study hours (simple regression)
from sklearn.linear_model import LinearRegression
hours = [[2], [4], [6], [8], [10]]
marks = [8, 10, 12, 14, 16]
model = LinearRegression().fit(hours, marks)
print(model.predict([[7]]))     # about 13
```

Worked examples

Example: A team builds an app that recognises Cameroonian banknotes for blind users. Suggest how they should collect and prepare the data.

> Photograph every banknote many times in different light, angles and conditions (old and new notes); label each photo with its value; remove blurred images; split into training and test sets.

Practice Questions: Lesson 5

1. Name two Python libraries used for machine learning.
2. Why is data split into training and test sets?

Answers: *1. E.g. scikit-learn, TensorFlow (PyTorch) 2. To measure performance on data the model has not seen*

Lesson 6: Integration activities

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply AI concepts to a community problem

Lesson notes

Situation: The council of a town wants to use AI to reduce traffic jams and to answer citizens' questions about birth certificates and permits.

Worked examples

Example: Task 1. Suggest one AI application for each need.

> Traffic: computer vision on cameras counts vehicles and an AI system adjusts traffic-light timings (reinforcement learning). Citizens: an NLP chatbot on WhatsApp or the council website answers questions in English and French.

Example: Task 2. List the data needed for the traffic system.

> Vehicle counts per road and time, junction layouts, current light timings, accident and event data.

Example: Task 3. Give three ethical precautions.

> Blur faces and number plates (privacy); keep humans able to override the lights (oversight); publish how the system works and its results (transparency).

Example: Task 4. The chatbot is right in 912 of 1 000 test questions. Find its accuracy and say what should happen to the 88 failures.

> 91.2%. The wrong answers should be studied, the training data improved, and difficult questions passed to a human officer.

Practice Questions: Lesson 6

1. Which AI technology would count vehicles from cameras?
2. How should the chatbot handle questions it cannot answer?

Answers: *1. Computer vision 2. Pass them to a human officer*

Lesson 7: Fundamentals of computer Networks

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define a computer network and its elements
- ✔ distinguish IP and MAC addresses

KEY POINTS

A **computer network** is two or more computers and other devices linked together to share data, resources and communication.

Advantages of networks	Disadvantages
Share files and data	Viruses and malware spread easily
Share hardware (printers, internet connection)	Security risks: hacking, data theft
Communication: e-mail, messaging, video calls	Cost of equipment, cabling and management
Central backups and administration	Failure of the server or network stops work
Software licences for many users	Needs skilled administrators

Basic elements of a network

Element	Role
Nodes (hosts)	Computers, phones, printers and servers connected to the network
Network interface card (NIC)	Connects a device to the network; has a unique MAC address
Transmission media	Cables (twisted pair, fibre) or wireless (radio, Wi-Fi)
Network devices	Switches, routers, access points, modems
Protocols	Rules for communication (TCP/IP, HTTP)
Network software	Network operating system, drivers, applications

KEY POINTS

Bandwidth is the maximum amount of data a link can carry per second (bps, Mbps, Gbps). **Latency** is the delay for data to travel from source to destination.

Every device on an IP network has an **IP address** (logical, e.g. 192.168.1.10) and a **MAC address** (physical, burnt into the NIC, e.g. 3C:52:82:4A:19:F0).

Example: Give three resources a school can share by networking its computers.

The internet connection, printers, and files such as mark sheets stored on a central server.

Practice Questions: Lesson 7

1. Define bandwidth.
2. Give one advantage of networking computers in an office.

Answers: *1. The maximum amount of data a link can carry per second 2. E.g. sharing printers, files and the internet connection*

Review Exercise: Applying Artificial Intelligence to improve on working and living standards

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. Define artificial intelligence and give three ways it improves living standards in Cameroon.
2. State and explain four principles of responsible AI.
3. Distinguish between forward chaining and backward chaining.
4. Describe the three types of machine learning with an example each.
5. A model gives 455 correct predictions out of 500. Find its accuracy.
6. List the stages of developing an AI system.
7. Define a computer network and give three advantages and two disadvantages.
8. Distinguish between an IP address and a MAC address.

Answers: *1. Systems that perform tasks requiring human intelligence; e.g. crop-disease detection, fraud detection in mobile money, AI tutors 2. Fairness, transparency, privacy, accountability (also safety, human oversight) 3. Forward: facts → rules → conclusions; backward: goal → find supporting facts 4. Supervised (labelled data, fraud detection), unsupervised (clusters, customer groups), reinforcement (rewards, robot) 5. 91% 6. Define the problem, collect data, prepare data, choose and train a model, evaluate, deploy, monitor 7. Linked devices sharing resources; advantages: sharing files, hardware and internet, communication, central backups; disadvantages: security risks, cost 8. IP address: logical address that can change, used for routing; MAC: physical address fixed in the network card*

END OF FREE SAMPLE

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